

Recap what is Internet? ^{a global} network of networks

◦ Addressing

* Physical address | MAC address (48-bit) ^{Medium access control}
• unique identification of each machine over a local network

48 bit address $\rightarrow$ Hexadecimal notation
(6B) eg. 1A:23:BC:DE:12:FF

... embedded on the NIC hardware
* Logical address eg IP address
• Uniquely identifies a machine over the Internet
IPv4: 32-bit (0 to $2^{32}-1$); IPv6: 128-bit address

* Port address: uniquely identifies a process on a machine
(16-bit) 0-1023: std. addr / public servers
1024-49151: semi-reserved; assigned by IANA
49152-65535: ephemeral port addr / client side

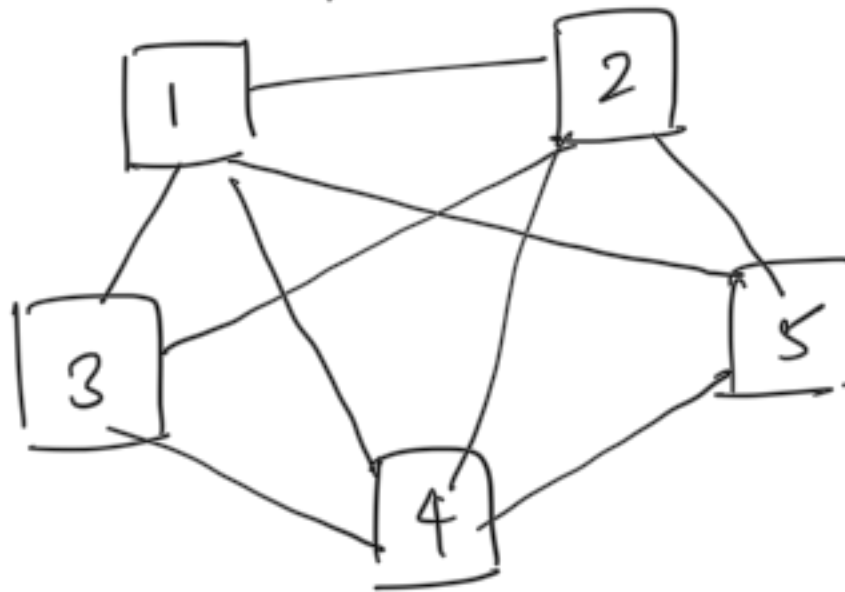
* Socket address $\Rightarrow$ IP addr, port addr
• Uniquely identifies a process on entire Internet

* Connection $\Rightarrow$ Two END POINTS $\Rightarrow$ src socket, dest socket
 $\downarrow$
4tuple $\Rightarrow$ (src IP, src port, dest IP, dest port)

* Network design requirements ① Scalability ...

- Network topologies

① Point-to-point



• dedicated path

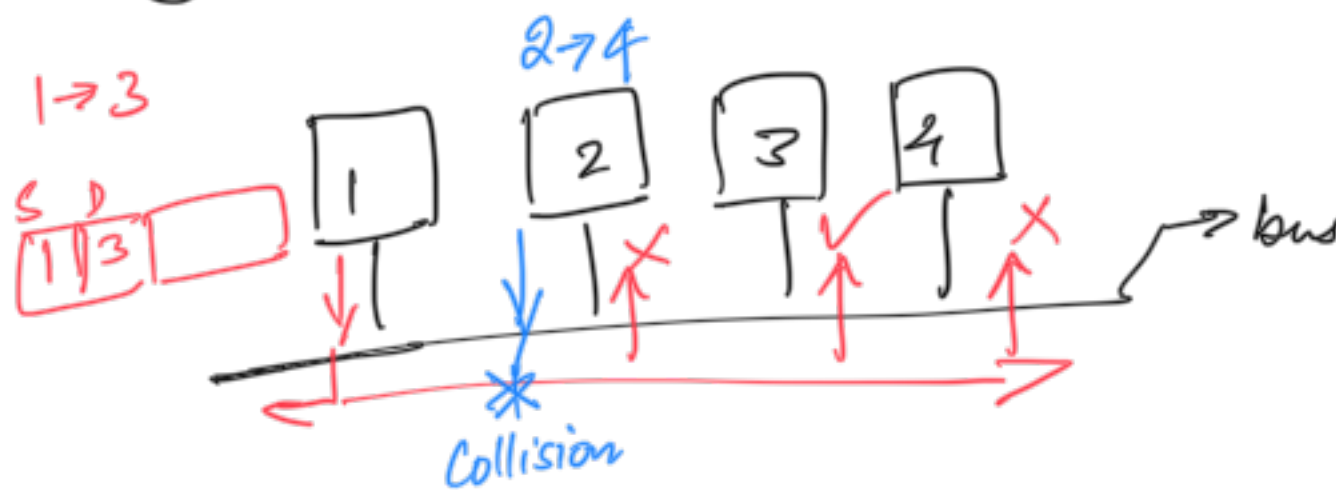
• robust

• fast, reliable

• does not scale

• $\frac{n(n-1)}{2}$ links

② Multipoint network (aka. Broadcast network)



• not robust

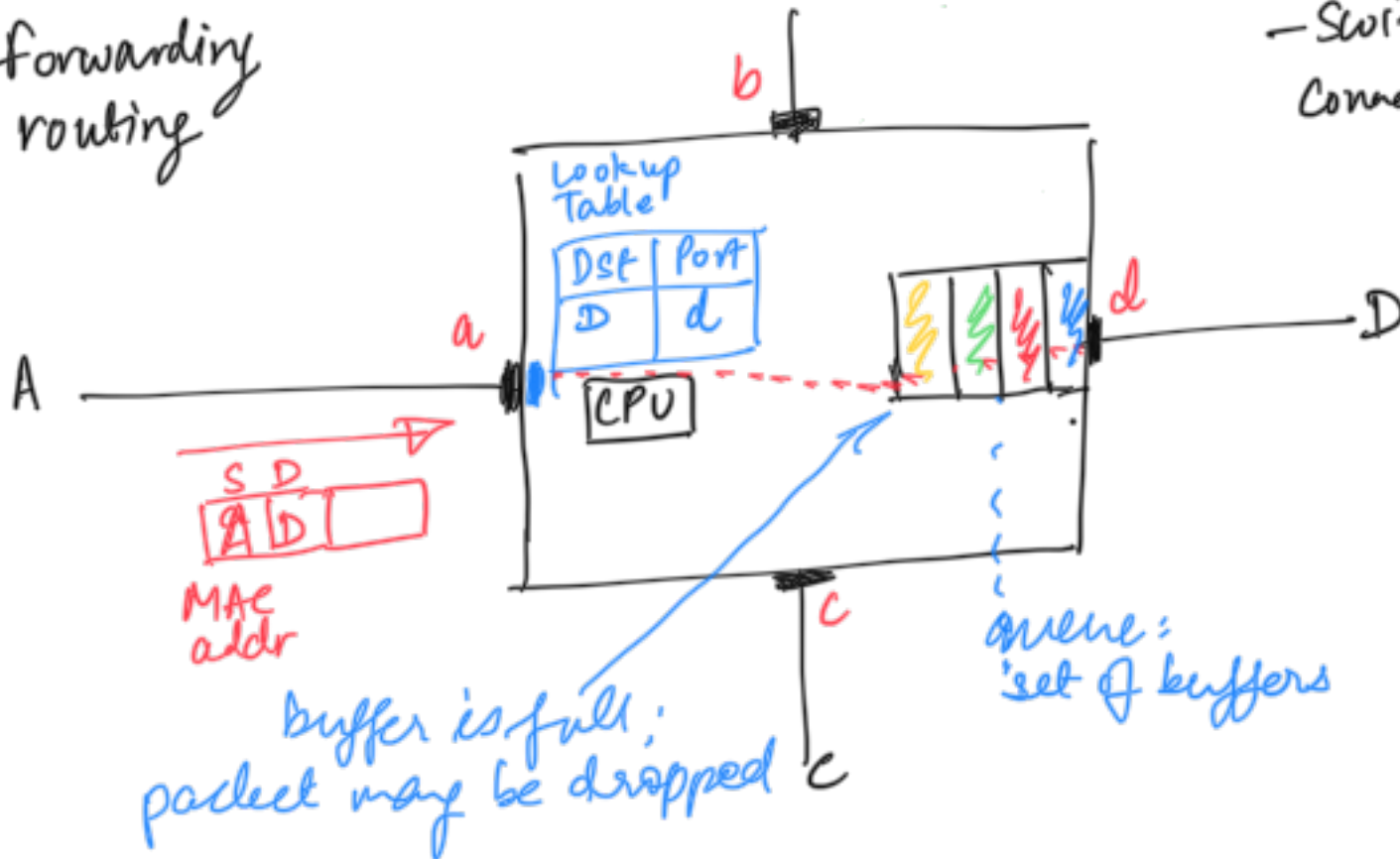
• scale

• low performance.

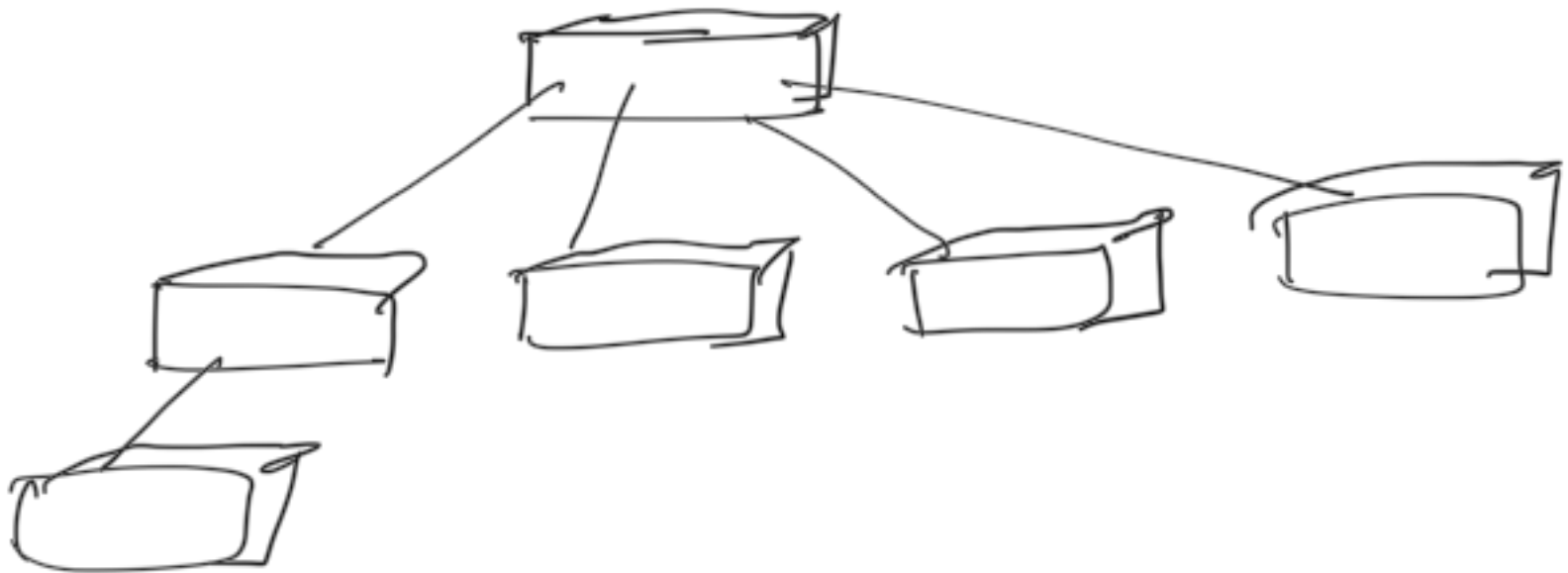
③ Switched network design

- forwarding
- routing

- Switching
Connect i/p to o/p

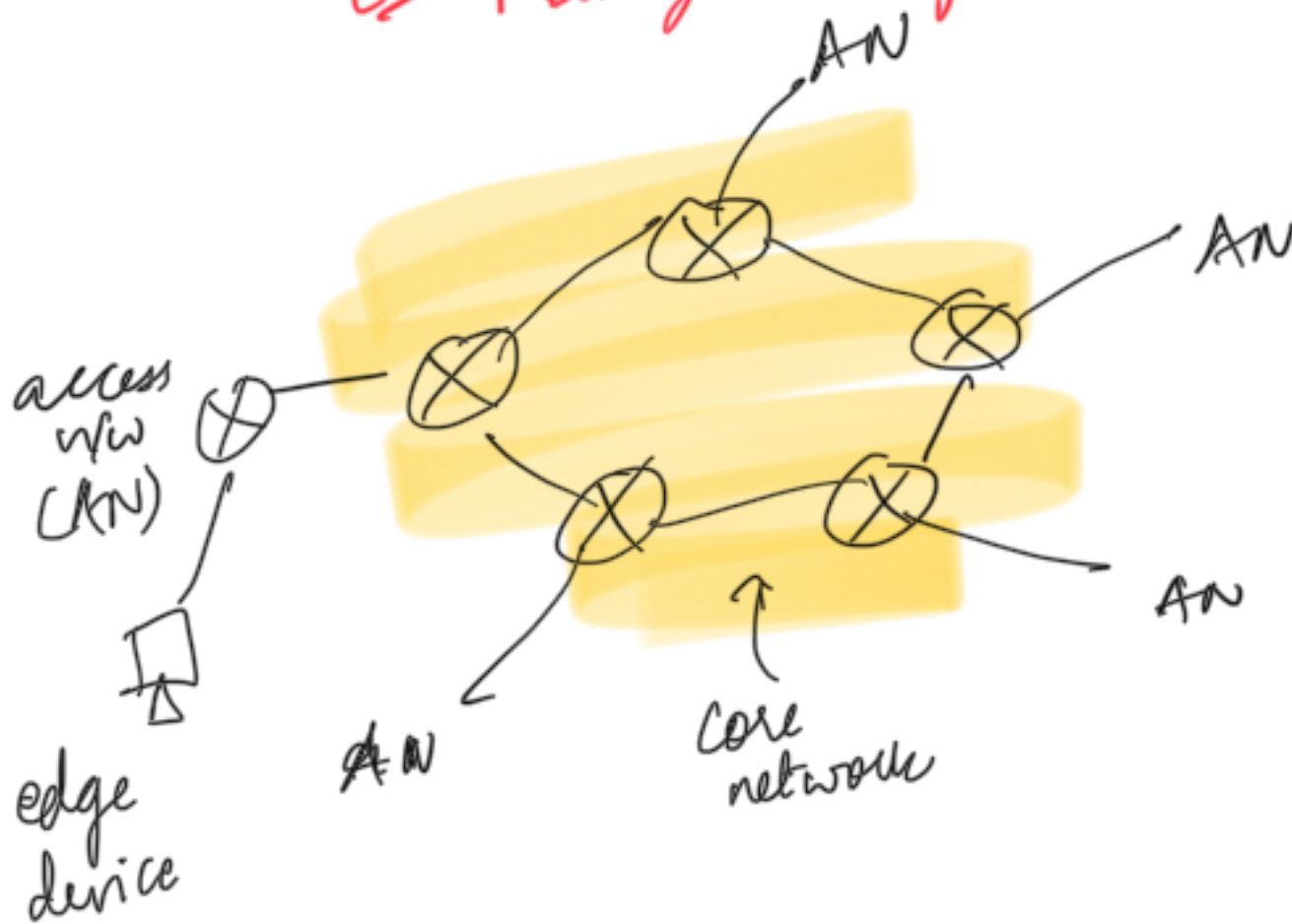


* Tree / Hybrid

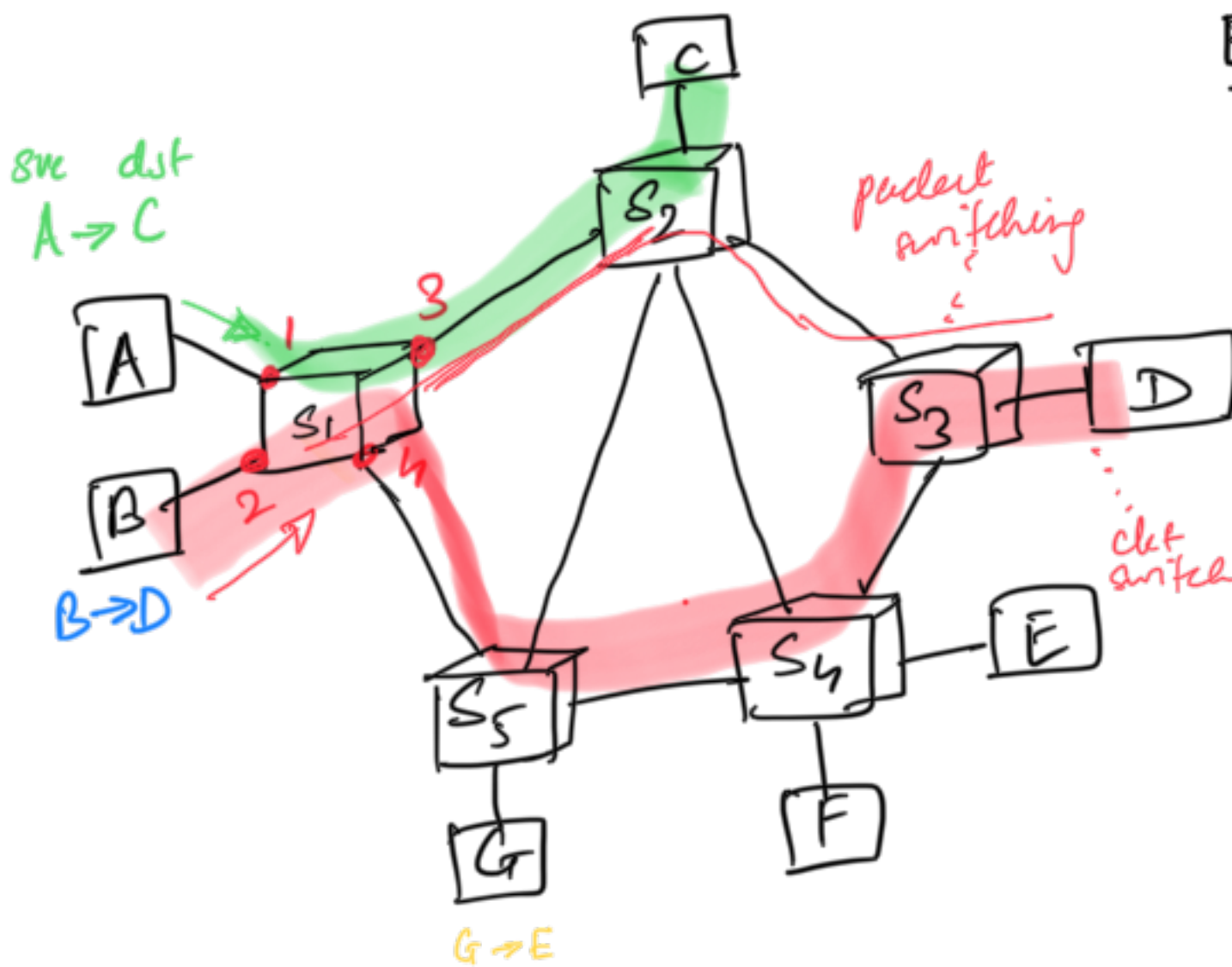


Network design requirements

- ↳ Scalability *
- ↳ Efficient resource utilization *
- ↳ Provisioning flow-specific services
- ↳ Manageability



* SWITCHING TECHNIQUES



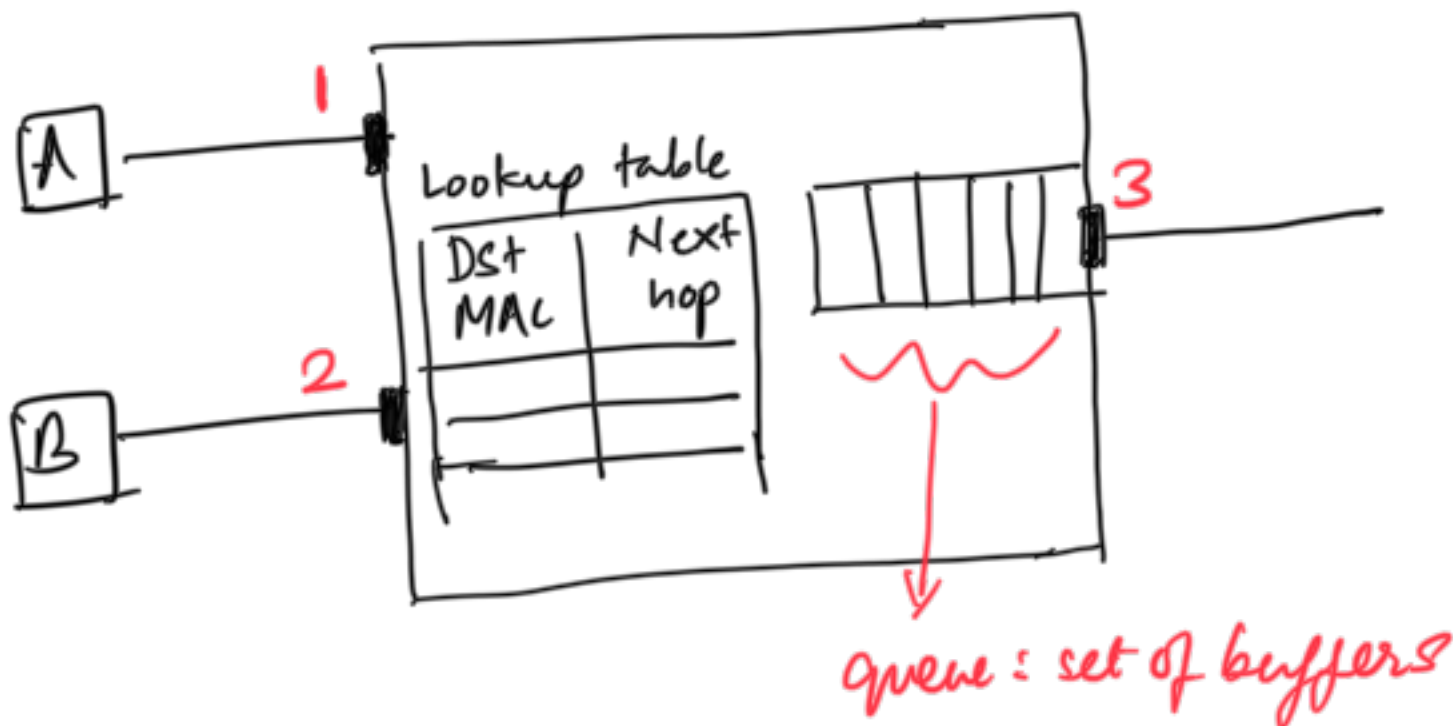
Example Connⁿ

- A → C
- B → D
- G → E

* Circuit switching
• dedicated + fixed path

* Packet switching
• path can be shared

Switch, S1



* Circuit switching vs. Packet switching

o Circuit switching



Call setup req
(A → C)

Propagation delay ' t_{prop} '

Processing delay ' t_{proc} '

Call setup process

* Connection oriented network service (CONS)

- connⁿ estab
- data xfer
- connⁿ termⁿ

processing delay can be
(a) zero; if resources on reverse path were reserved during call setup

(b) non-zero otherwise

DATA

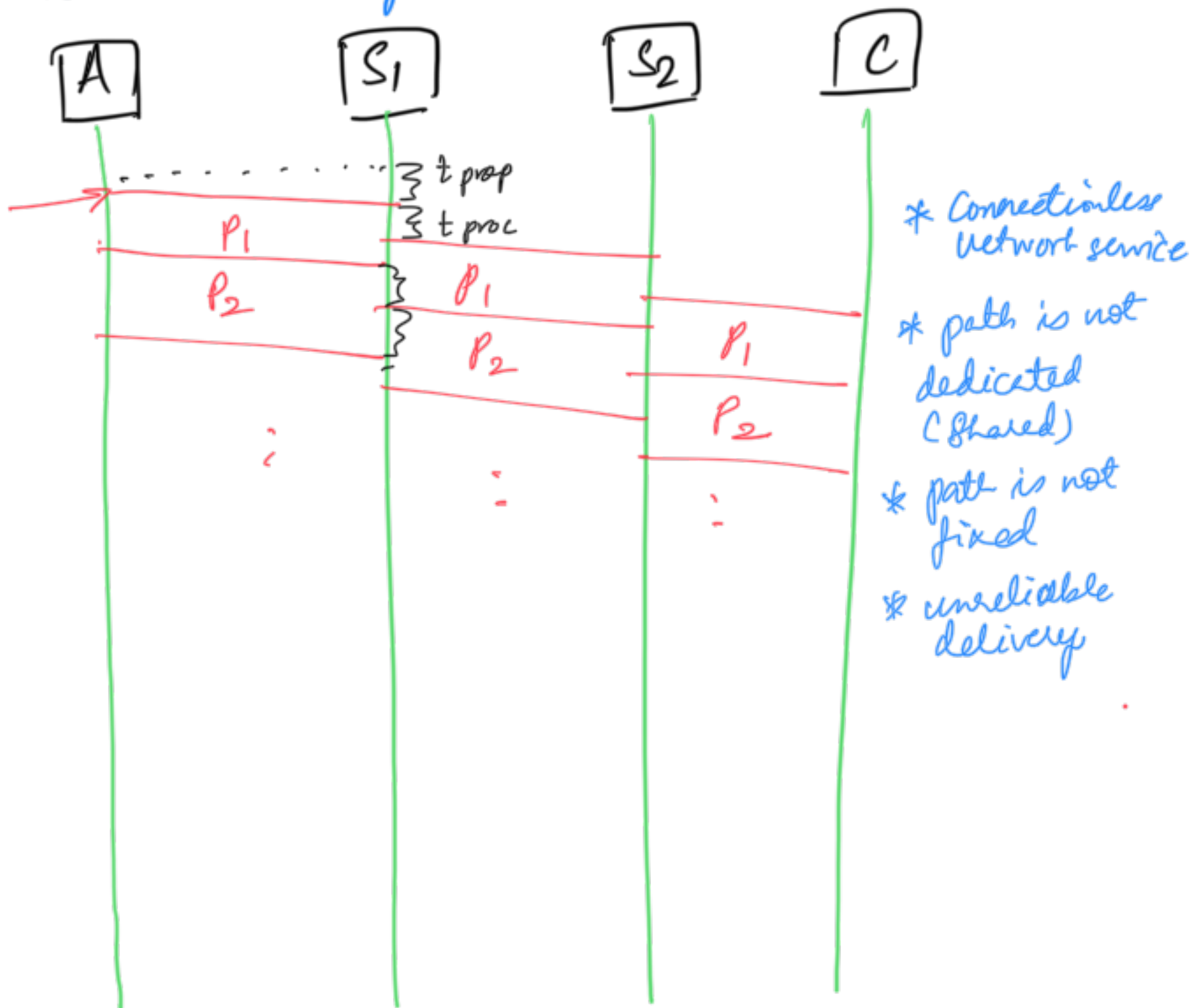
DATA

Call termination request

Ack

Resources released

* Packet switching -- Datagram switching



- ## Circuit switching
- slow for short data
 - reliable
 - reduced overhead in terms of pkt headers
 - low latency

- ## Packet switching
- fast for short data
 - unreliable
 - high overhead
 - high latency
 - BW is shared

HOMEWORK

